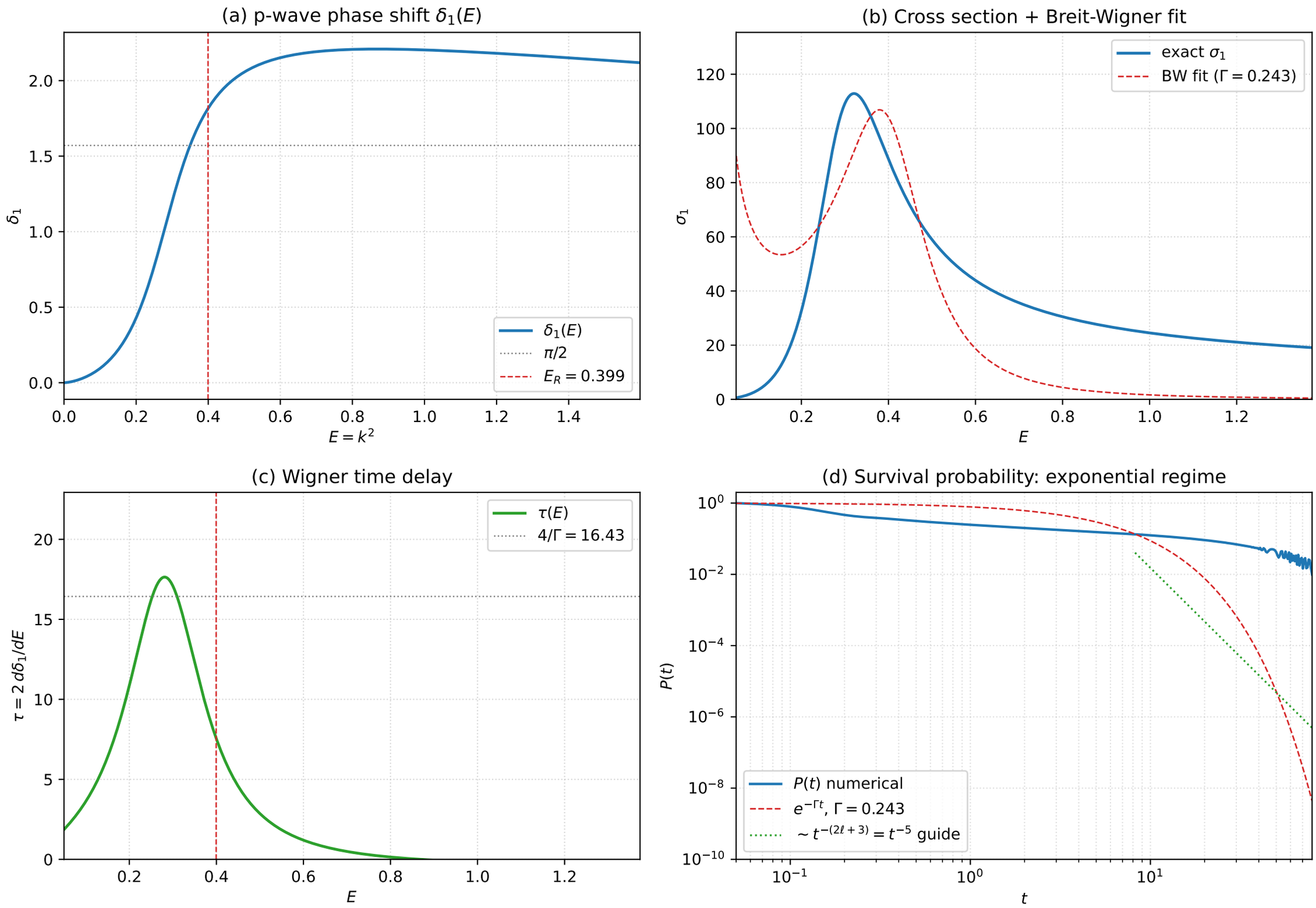
(a) p-wave phase shift $\delta_1(E)$
δ_1
$E = k^2$
$\delta_1(E)$
$\pi/2$
$E_R = 0.399$
(b) Cross section + Breit-Wigner fit
σ_1
E
exact σ_1
BW fit ($\Gamma = 0.243$)
(c) Wigner time delay
$\tau = 2\,d\delta_1/dE$
E
$\tau(E)$
$4/\Gamma = 16.43$
(d) Survival probability: exponential regime
$P(t)$
t
$P(t)$ numerical
$e^{-\Gamma t}$, $\Gamma = 0.243$
$\sim t^{-(2\ell+3)} = t^{-5}$ guide